MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6418

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, subminiature, voltage amplifier
Outline --- See figure 4
Cathode --- Coated filament
Base connections:

Pin No. ---	1	2	3	4	5
Element ---	a (Red dot)	g2	+f, g3 (Note 1)	g1	-f, g3 (Note 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik	Rg1	(D) Alt
Unit:	Vdc	Vdc	Vdc	Vdc	mA	Meg	ft
Maximum:	1.50	30	---	30	0.50	---	Note 3
Minimum:	1.00	---	---	---	---	---	---
TEST CONDITIONS:	1.25 (Note 2)	22.5	-1.2	22.5	---	0.5 (max)	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1266	Total grid current	-		0.65	II	Ic1	0	-0.5	μAdc
1256	Electrode current (1) (anode)	-		0.65	II	Ib	150	300	μAdc
1256	Electrode current (1) (screen)	-		0.65	II	Ic2	---	85	μAdc
1341	Power output (1)	-	Esig = 0.85 Vac; Rg1 = 5.0 Meg; Rp = 0.1 Meg	0.65	II	Po	1.4	---	mW
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Filament current	-		---	---	If	8.8	11.2	mA
1266	Grid emission	-	Ec1 = -50 Vdc	---	---	Ic1	0	1.0	μAdc
1341	Power output (2)	-	Ef = 1.0 Vdc; Esig = 0.85 Vac; Rg1 = 5.0 Meg; Rp = 0.1 Meg	---	---	Po	1.2	---	mW
1306	Transconductance	-		---	---	Sm	200	400	μmhos
1311	Anode resistance	-		---	---	Rp	0.3	---	Meg
1316	Amplification factor	4		---	---	Mu	6.5	9.0	---
1256	Electrode current (2) (anode)	5		0.65	I	Ib	235	---	μAdc
1256	Electrode current (2) (screen)	5		0.65	I	Ic2	95	118	μAdc
1256	Electrode current (3) (anode)	6		0.65	I	Ib	---	15	μAdc
1256	Electrode current (3) (screen)	6		0.65	I	Ic2	---	6	μAdc
---	Operation tap	7		1.0	I	---	---	---	---
1116	Lead fatigue	-		4.0	S3	---	---	---	---
(D) 1119	Glass-to-lead seal strain	-		---	---	---	---	---	---
(D) 1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL & CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3									
(D) 1521	Survival-rate life	See figure 3	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points:									
1256	Electrode current (2) (anode)		---	---	---	---	Ib	225	---	μAdc
1256	Electrode current (3) (anode)		---	---	---	---	Ib	---	16	μAdc
1256	Electrode current (2) (screen)		---	---	---	---	Ic2	90	---	μAdc
1256	Electrode current (3) (screen)		---	---	---	---	Ic2	---	9	μAdc
---	Inoperatives		---	---	---	---	---	---	---	---
1501	Intermittent life	See figure 3	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic1	---	-0.5	μAdc
1341	Power output (1)		---	---	1	3	Po	1.0	---	mW
1256	Electrode current (2) (anode)		---	---	1	3	Ib	220	---	μAdc
1256	Electrode current (2) (screen)		---	---	1	3	Ic2	75	---	μAdc
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points (1,000 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic1	---	-0.5	μAdc
1341	Power output (1)		---	---	1	3	Po	0.9	---	mW
1256	Electrode current (2) (anode)		---	---	1	3	Ib	200	---	μAdc
1256	Electrode current (2) (screen)		---	---	1	3	Ic2	70	---	μAdc
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. Grid No. 3 is composed of two separate deflector plates, one of which is connected to pin No. 3 and the other to pin No. 5.
2. Do not use filaments in series.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
4. Tie grid No. 2 to anode.
5. See circuit No. 1, S1 open as shown on figure 1.
6. See circuit No. 1, S1 closed as shown on figure 1.

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NOTES: -Continued

7. This test shall be conducted in the bistable multivibration circuit No. 2 shown on figure 2. At the start of the test the NCT (normally conducting tube) shall be conducting and the TUT (tube under test) shall be cut off. The TUT shall be tapped three times in each of two directions mutually perpendicular. The tapper used shall be the automatic type using G. E. Teflon tipped hammers, or equivalent. The amplitude and duration of each tap shall be 80G ±10 percent and 250 ±50 μs, when measured using a suitable accelerometer and output measuring device. If a short within the TUT occurs there will be a complete change of state in the circuit. The TUT will begin to conduct and will not be cut off. The cause of reject shall be when the NCT is cutoff which will be indicated by a decrease in current through the conducting indicator (a meter). The reset microswitch, shown in circuit No. 2, when depressed will throw the 5 megohms into the NCT grid circuit thereby reducing the bias. This will again cause the NCT to draw current and the circuit will be reset.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2608)

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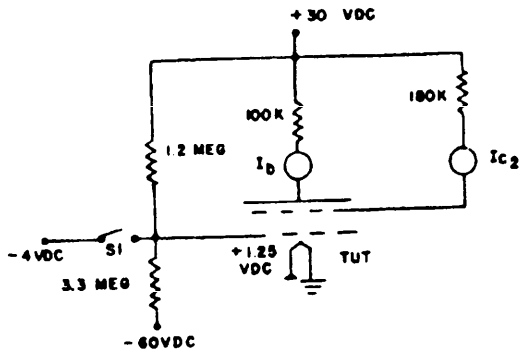


FIGURE 1. Circuit No. 1.

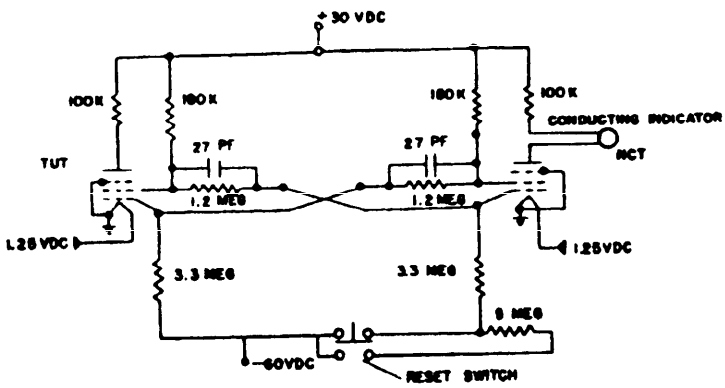


FIGURE 2. Circuit No. 2.

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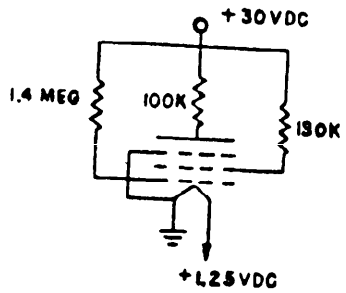
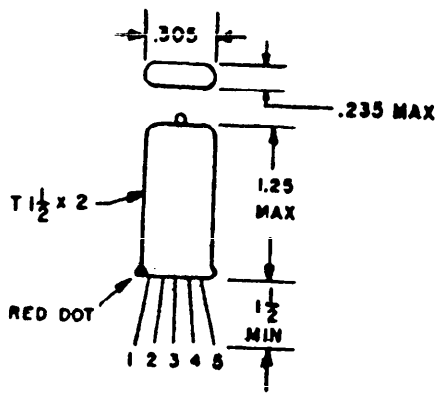


FIGURE 3. Circuit No. 3.



Base: Pinch press (0.016-inch tinned flexible leads spaced 0.040 inch c/c).

FIGURE 4. Outline drawing of electron tube type 6418.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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